The Sun Is Not a Ruler: What Cast Shadows Can and Cannot Check in Monocular Depth

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Abstract

GPS and a timestamp give the exact direction of the sun, so a correct depth map should cast the shadows a camera actually sees. This makes cast shadows an attractive label-free check for monocular metric depth. We show what such a check can and cannot measure. In theory, global metric scale is invisible: a scaled scene casts the same image shadows. A wrong focal length is exactly *a wrong sun direction, and a local depth warp is one to first order. The equivalent sun error has a closed-form bound, 14.5° for a 40% focal error and 6.4° for 20%. Compressed depth therefore acts like a lower sun and casts longer shadows. On an exact synthetic street the check works for shape and focal length and is blind to scale, as predicted. On 71 KITTI frames with four metric depth models and a learned shadow detector (ISTD BER 1.9%), it fails. The shadow IoU is highest for the most compressed depth, the true value is the per-frame best only 5–10% of the time (chance 20%), and the score ranks DA-V2-L below DA-V2-S. The reason is coverage. IoU rewards the larger rendered shadow, and at the true depth the agreement is at chance level (IoU 0.064 against 0.062 for independent masks, $\kappa = -0.003$). The corruption protocol is a test that any future shadow-based depth signal should pass.*

1. Introduction

Monocular metric depth models such as Depth Anything V2 [30], Depth Pro [2], MoGe-2 [27], Metric3D v2 [15] and UniDepth [23] now give plausible metric depth from one image. Deployed on a new camera or a new city, however, they are used without ground truth. A check that needs no labels would let a user see when the depth is wrong.

Outdoors, physics offers one for free. A photo with GPS and a timestamp fixes the sun direction to a fraction of a degree [14, 25]. If the depth map and camera are right, rendering cast shadows from them under that sun should reproduce the dark regions in the image. Shadows have been used to recover shape [1, 8, 19], camera parameters [3, 21] and location [18]. Using them to *check* a learned depth map is the natural next step.

This paper asks when that check can work, and answers in theory and in experiment.

What a shadow can see. Because the sun is at infinity, scaling the scene about the camera leaves every image-space shadow unchanged (Proposition 1). The quantity the check was meant for, metric scale, is unobservable. A wrong focal length is not invisible, but it is *exactly* equivalent to a wrong sun direction (Proposition 2), and a local shape warp is equivalent to one to first order. The equivalent sun error has a closed-form worst case (Corollary 1): 14.5° for a 40% focal error, 6.4° for 20%. Compressed depth always acts like a lower sun and casts longer shadows (Corollary 2). A shadow check can therefore test shape and intrinsics only as well as it can locate the sun, from the image, to a few degrees.

What it sees in practice. On an exact synthetic street the check behaves as the theory says: blind to scale, and clearly peaked near the true shape and focal length (Sec. 5.1). On real KITTI drives [10] it fails, and it fails in a specific way (Fig. 1). With four current metric depth models, a threshold detector and a SegFormer [29] shadow detector that we train on ISTD [26], the score rises steadily as depth is compressed. The true value is the best one in only 5% of cases for focal length (chance 20%). The score also ranks a weaker model above a stronger one (Sec. 5.2).

Why. Compression lengthens the rendered shadows (Corollary 2). IoU against a detector that marks about a third of the road rewards the larger mask even when it is in the wrong place. At the true depth the rendered and observed shadows agree no better than two independent masks of the same coverage, and a chance-corrected score [6] drops to zero (Sec. 5.3). The failure is not in the scoring

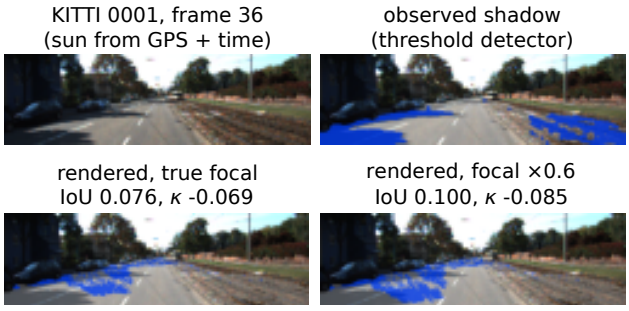


Figure 1. A KITTI frame with the sun from GPS and time. Shadows rendered from DA-V2-S depth with the true focal length (bottom left) and with a focal length 0.6 times too short (bottom right), against the observed road shadow (top right). The wrong camera casts longer shadows and scores the higher IoU. The chance-corrected κ is below zero for both.

rule. On real frames the rendered shadows simply do not carry the information.

Contributions.

- An identifiability analysis of shadow-based depth checking: scale invariance, the exact equivalence of focal and sun errors, its first-order extension to shape warps, a closed-form bound on the equivalent sun error, and the prediction that compressed depth casts longer shadows.
- A controlled corruption protocol (scale, shape, focal, sun) with per-frame "truth is argmax" tests and drive-level bootstraps. Any proposed shadow score can be run through it.
- A synthetic confirmation and a negative result on real driving data with four metric depth models and a learned shadow detector, explained by a coverage bias and chance-level agreement.

Every number in the paper is generated directly from the result files.

2. Related work

Geometry from shadows and the sun. Shape from shadows recovers surface profiles from moving light sources [8, 19]. The episolar constraint [1] uses the known sun path over a day of webcam images to obtain sparse depth from shadow correspondences. Solar shadows also calibrate cameras and give light direction [3], and shadow trajectories give geo-temporal location [18]. Lalonde *et al*. [20, 21] and Hold-Geoffroy *et al*. [13] estimate the sun and camera from sky appearance and shadows. In multi-view and per-scene reconstruction, shadows are a training signal: S-NeRF [9] models the known sun in satellite images, ShadowNeuS [22] reconstructs an SDF from shadow rays, and OutCast [12] relights single images with cast shadows from estimated depth. Closest to us, Hu *et al*. render shadows from MoGe-2 point maps and an estimated light direction, then refine them with a diffusion model [16]. They also feed light and normals back into attached-shadow detection [17]. These works use many images or a moving light, fit geometry to the shadows, or trust the depth to make shadows. We ask the converse question for one image and a fixed sun: whether shadows can *test* a given depth map, and which of its errors they can see. Our scale invariance is the single-image form of the projective ambiguity these methods resolve with extra views or known sizes.

Shadow detection. Learned detectors [4, 28, 32] are trained on ISTD [26] and SBU-style data, mostly close-up scenes. We train a SegFormer [29] on ISTD and compare it with a dark-asphalt threshold. The detector is not the bottleneck: both detectors lead to the same conclusion.

Label-free signals for depth. Self-supervised depth uses photometric consistency across frames [11, 31], and depth uncertainty can be estimated without labels [24]. These need video or model internals. A sun-shadow check would need neither, which is why its limits are worth knowing.

3. What a cast shadow constrains

Setup. We use camera coordinates with x right, y down and z forward. A depth map d and intrinsics K lift pixel u to $X(u) = d(u)\,K^{-1}\tilde{u}$, and the lifted points form a surface S. The sun is at infinity, with unit direction s pointing towards it. Pixel u is in shadow when the ray $X(u) + t\,s$, $t > 0$, meets an occluder. The occluder is S, thickened behind its visible side in practice. We write $M(d, K, s)$ for the resulting binary image, the *rendered shadow*. A shadow check compares M with an observed shadow mask O through a score $\phi(M, O)$.

Proposition 1 (Scale is invisible)**.** *For every* $\lambda > 0$, $M(\lambda d, K, s) = M(d, K, s)$.

Proof. Scaling d maps $X(u)$ to $\lambda X(u)$ and S to λS. Then $\lambda X + ts \in \lambda S$ if and only if $X + (t/\lambda)s \in S$, and t/λ ranges over all $t > 0$. □

Any score of M is therefore constant in the global scale. This is the error metric depth models make most often, and the one a label-free check was meant to catch. A renderer with tolerances in metres (self-shadow bias, occluder thickness) breaks the invariance slightly, and scaling the tolerances with the scene restores it exactly (Sec. 5.1).

Proposition 2 (A focal error is a sun error)**.** *Let* K_k *be* K *with both focal lengths multiplied by* $k > 0$ *and the princi-*

pal point kept, and let $D = \mathrm{diag}(1/k, 1/k, 1)$. *Then*

$$M(d, K_k, s) = M\big(d, K,\, s_k\big), \qquad s_k = \frac{D^{-1}s}{\|D^{-1}s\|} \propto (ks_x,\, ks_y,\, s_z). \tag{1}$$

Proof. With K_k the same depth lifts u to $DX(u)$, so the surface is DS. Since D is linear and invertible, $DX + ts \in DS$ if and only if $X + t\,D^{-1}s \in S$. Normalising the direction only rescales t. □

The rendered shadow from a wrong camera is *exactly* the true scene's shadow under a different sun. The same holds with separate factors on f_x and f_y. Proposition 1 is the special case $D = \lambda I$, which leaves s unchanged. Table 1 and Fig. 2 show why the focal and shape curves look the same: a depth warp $d \mapsto m(d/m)^\gamma$ about the median m stretches the scene locally by γ along the line of sight relative to the other directions. Near the image centre it therefore acts like $D = \rho\,\mathrm{diag}(1, 1, \gamma)$ for a local scale ρ. Its equivalent sun $(s_x, s_y, s_z/\gamma)$ has the direction of s_k with $k = \gamma$. This holds to first order only, since a caster and its shadow are a finite distance apart.

Corollary 1 (Bound on the equivalent sun error)**.** *Let* ψ *be the angle between* s *and the optical axis. The angle between* s *and* s_k *is* $|\arctan(k\tan\psi) - \psi|$. *Its maximum over* ψ *is*

$$\theta_{\max}(k) = \arctan\frac{|1-k|}{2\sqrt{k}}, \tag{2}$$

reached at $\tan\psi = 1/\sqrt{k}$. *The error is zero when the sun lies on the optical axis or in the image plane (*$\psi = 0$ *or* $90°$*).*

Proof. D^{-1} scales the component of s orthogonal to the axis by k and leaves the axial one, so $\tan\psi' = k\tan\psi$. With $t = \tan\psi$, $\tan(\psi - \psi') = t(1-k)/(1+kt^2)$, which is maximal at $t = 1/\sqrt{k}$ with value $(1-k)/(2\sqrt{k})$. □

Equation (2) gives 6.4° for $k = 0.8$ or 1.25, and 14.5° for $k = 0.6$. For the sun of KITTI drive 0001 the actual equivalent errors are smaller: 3.5° at $k = 0.8$ and 8.9° at $k = 0.6$ (Fig. 4, right). A shadow check can detect a 20% focal or shape error only if it can tell the true sun from one a few degrees away, in a single image. The sun direction itself is known that well, from GPS, time and the vehicle pose. The question is whether the observed shadows constrain it that tightly.

Corollary 2 (Compressed depth is a lower sun)**.** *Let the ground normal be* $-y$, *so the sun elevation* ϵ *satisfies* $\tan\epsilon = -s_y/\sqrt{s_x^2 + s_z^2}$. *For* $k < 1$ *the equivalent sun* s_k *is lower than* s, *and for* $k > 1$ *it is higher.*

Proof. $\tan\epsilon_k/\tan\epsilon = k\sqrt{s_x^2 + s_z^2}/\sqrt{k^2 s_x^2 + s_z^2}$, which is below 1 exactly when $k < 1$. □

A lower sun casts longer shadows on the ground. So a too-short focal length, or depth compressed towards its median, *enlarges* the rendered shadow. A score that grows with the overlap of a larger mask will then prefer the wrong depth. Section 5.3 shows that this is what happens on KITTI.

4. A shadow check and how to test it

Sun. The sun's azimuth and elevation come from the solar position algorithm [25] in pvlib [14], at the GPS position and time of each frame. They are rotated into the rectified left colour camera with the OXTS roll, pitch and yaw (KITTI devkit convention) and the IMU–LiDAR–camera calibration. KITTI timestamps are local time: shadows of poles in drive 0001 line up with the sun for UTC+2 (azimuth 175°, elevation 40°) and not for UTC. All frames used here were recorded on 26 September 2011, with sun elevation 39.7–39.8°.

Receivers and observed shadows. We score only the road, which is flat and close to albedo-neutral. The road is segmented with Mask2Former Swin-L [5] trained on Cityscapes [7]; the drive-0001 analysis in Sec. 5.3 uses SegFormer-B0. Two shadow detectors give the observed mask O. The *threshold* detector marks road pixels darker than 0.55 times the 80th percentile of road luminance, followed by a small opening and a median filter. The *learned* detector is a SegFormer MiT-B2 [29] trained for 8 epochs on the 1,330 ISTD [26] training images. Its balanced error rate on the 540 ISTD test images is 1.9%, in the range of published ISTD detectors. On the KITTI road the two detectors mark 27% and 33% of the pixels as shadow.

Rendering. Depth is predicted at full resolution and evaluated at a quarter (about 93×310). Sky pixels are removed. The rendered shadow M is computed in screen space: each pixel is lifted to 3D and marched towards the sun in 64 steps up to 25 m. It is in shadow if a step lands behind the visible surface by more than $0.15\,\mathrm{m} + 3\%$ of the depth and less than an assumed occluder thickness of 3 m. Screen space means that casters outside the image cannot cast shadows. LiDAR does not help here. In a pilot on drive 0001, LiDAR depth reached road IoU 0.01–0.10 against 0.06–0.30 for predicted depth, because tree canopies and facades lie above the LiDAR's vertical field of view.

Scores. We use the IoU of M and O on the road. We also use Cohen's κ, the pixel agreement corrected for the agreement two independent masks of the same coverage would reach [6]. For coverages r and o of M and O, independent masks have expected IoU $ro/(r + o - ro)$, which we report as the *chance IoU*.

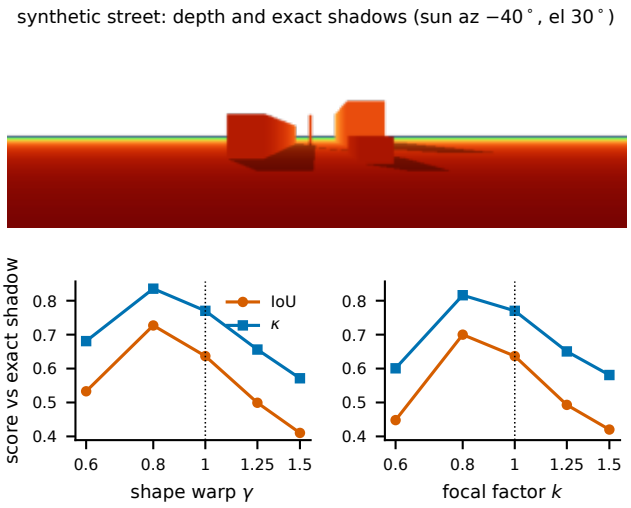


Figure 2. Synthetic street. Top: depth and exact shadows. Bottom: IoU and κ of the rendered shadow against the exact one, averaged over four suns, under a shape warp and a focal error. The two corruptions give almost the same curves, as Proposition 2 predicts.

Depth models. DA-V2-S and DA-V2-L (Metric Outdoor) [30] predict depth from pixels alone. Depth Pro [2] and MoGe-2-L [27] also predict the camera. For rendering we use KITTI's calibrated K for all four, so that only the depth differs.

Corruption protocol. A useful check must score the *true* geometry best. We corrupt each predicted depth map in one factor at a time, with five values each: global scale $d \mapsto \lambda d$ ($\lambda \in \{0.5, 0.75, 1, 1.5, 2\}$); shape warp $d \mapsto m(d/m)^\gamma$ about the median m ($\gamma \in \{0.6, 0.8, 1, 1.25, 1.5\}$); focal factor $f \mapsto kf$ ($k \in \{0.6, 0.8, 1, 1.25, 1.5\}$); and sun azimuth rotated by $\{-60, -30, 0, 30, 60\}^\circ$ as a positive control. Each (frame, model) pair is a *case*. For each case and corruption we record whether the true value has the highest score (*truth is argmax*; chance is 20%). We also record the paired gain of the most compressed value over the truth. Frames of one drive are correlated, so intervals are 95% bootstraps over drives (2,000 resamples, each drive weighted by its number of cases).

5. Experiments

5.1. Synthetic street: the theory holds

We first remove every source of noise except the geometry. A synthetic street (ground plane, wall, parked car, pole and building; Fig. 2) is seen by a KITTI-like camera. Its true shadows come from exact 3D ray casting under four suns (elevations 25–45°), and they serve as the observed mask. Shadows are rendered from corrupted versions of the true depth with the screen-space renderer of Sec. 4.

Scale. With the renderer's metric tolerances scaled along with the scene, the IoU is 0.637 at every scale from 0.5 to 2, as Proposition 1 requires. With fixed tolerances it drifts slightly (0.666 at $\lambda = 2$), which is a renderer artefact and not a signal.

Shape and focal. Both corruptions are clearly visible. IoU falls from 0.637 at the truth to 0.448 and 0.420 at $k = 0.6$ and 1.5, and κ from 0.770 to 0.601 and 0.581. The shape and focal curves nearly coincide, the first-order equivalence of Sec. 3. Both peak at 0.8 rather than 1. The screen-space renderer misses about 13% of the exact shadow, because rays leave the image and thin casters are skipped. Compressed depth, which casts longer shadows (Corollary 2), makes up part of the gap. So with exact observed shadows the check sees shape and the camera, and it is blind only to scale.

5.2. KITTI: the check fails

We take every tenth frame of five KITTI raw drives from 26 September 2011 (0001, 0005, 0009, 0011, 0013; at most 15 frames per drive). That gives 71 frames, four depth models, two detectors and the four corruptions, 11,360 scored renderings in total (Tab. 1, Fig. 3).

The positive control passes, on average. Rotating the sun lowers the mean IoU for every model and both detectors, from 0.116 at the true sun to 0.071 at -60° (learned detector). The sun pipeline, the timestamps and the frames are therefore right. Per frame, however, the true sun is the best of the five directions in only 26% of cases [9, 46] (chance 20%). A single frame does not locate the sun to 30°. By Corollary 1, every focal error with k between $1/3$ and 3 is equivalent to a sun error below 30°, so a single frame cannot be expected to see any of the focal errors we test.

Scale is flat, as Proposition 1 predicts. Halving the depth changes the learned detector's IoU by -0.004 [-0.017, 0.010].

Shape and focal move the score the wrong way. The IoU rises steadily as depth is compressed, for all four models and both detectors. With the learned detector it rises from 0.116 at the true focal to 0.158 at $k = 0.6$. Per frame the truth is the best value in only 5% [3, 8] of cases for focal and 10% [1, 20] for shape, at or below the 20% of chance. A 40% too-short focal length scores 0.042 IoU better than the truth on average. With five drives the interval [-0.002, 0.105] just includes zero. With the threshold detector the gain is 0.021 [-0.004, 0.048].

	mean road IoU at corruption value					truth is argmax	IoU(wrong)−IoU(truth)
shadow detector: threshold							
scale λ (0.5 / 0.75 / 1 / 1.5 / 2)	0.074	0.081	0.085	0.088	**0.088**	7% [4, 12]	−0.010 [−0.021, +0.001]
shape γ (0.6 / 0.8 / 1 / 1.25 / 1.5)	**0.103**	0.092	0.085	0.079	0.076	5% [4, 7]	+0.019 [−0.005, +0.055]
focal k (0.6 / 0.8 / 1 / 1.25 / 1.5)	**0.106**	0.093	0.085	0.078	0.075	6% [3, 11]	+0.021 [−0.004, +0.048]
sun az. (deg) (-60 / -30 / 0 / 30 / 60)	0.059	0.081	**0.085**	0.065	0.055	23% [8, 40]	−0.026 [−0.057, +0.005]
shadow detector: SegFormer-ISTD							
scale λ (0.5 / 0.75 / 1 / 1.5 / 2)	0.112	**0.116**	0.116	0.113	0.107	10% [6, 15]	−0.004 [−0.017, +0.010]
shape γ (0.6 / 0.8 / 1 / 1.25 / 1.5)	**0.148**	0.131	0.116	0.107	0.100	10% [1, 20]	+0.033 [−0.004, +0.090]
focal k (0.6 / 0.8 / 1 / 1.25 / 1.5)	**0.158**	0.133	0.116	0.106	0.101	5% [3, 8]	+0.042 [−0.002, +0.105]
sun az. (deg) (-60 / -30 / 0 / 30 / 60)	0.071	0.104	**0.116**	0.095	0.072	26% [9, 46]	−0.045 [−0.082, −0.000]

Table 1. The shadow score on KITTI (71 frames from 5 drives, four depth models). Columns are the five corruption values in the order given in brackets; the true value is underlined, the best mean in bold. 'Truth is argmax' is the share of (frame, model) cases whose best value is the true one (chance 20%), and the last column the paired IoU gain of the most compressed value (scale 0.5, γ and k 0.6, azimuth $-60°$) over the truth; intervals are 95% bootstraps over drives.

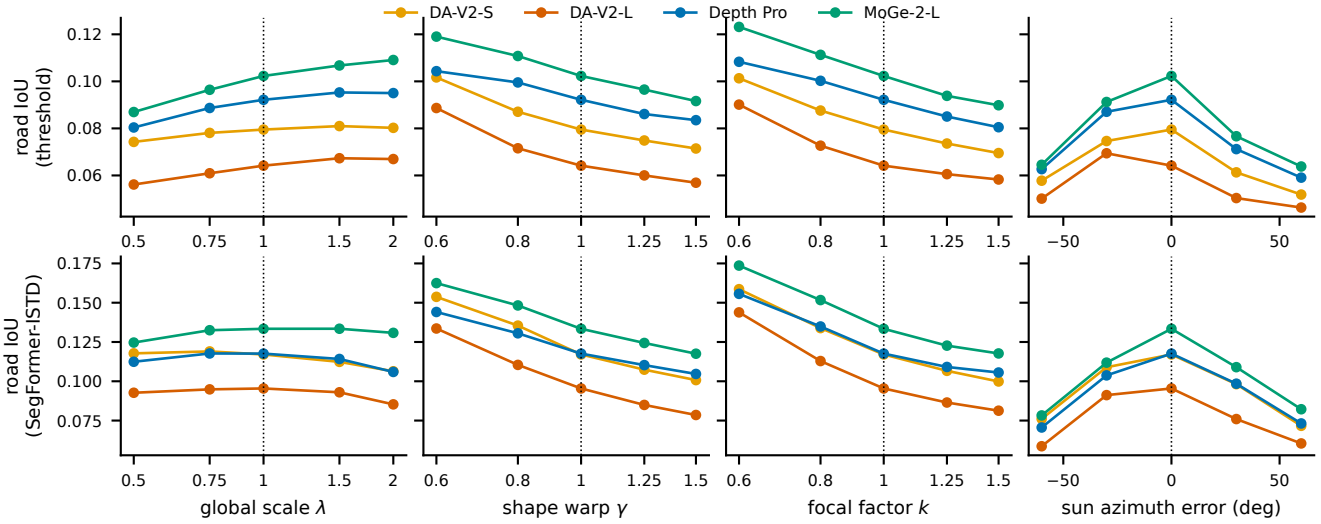


Figure 3. Mean road shadow IoU on KITTI per depth model, for both detectors and all four corruptions (dotted: true value). The sun control peaks at the truth. Scale is flat. Shape and focal rise steadily towards the most compressed value.

A better detector does not help. The ISTD-trained detector raises the IoU level (about 0.12 against 0.08 at the truth) but keeps the same trend.

The score does not rank models. On KITTI, DA-V2-L is more accurate than DA-V2-S [30]. The shadow score puts it last with both detectors (Tab. 2). MoGe-2-L scores highest.

detector	DA-V2-S	DA-V2-L	Depth Pro	MoGe-2-L
threshold	0.080	0.064	0.092	0.102
SegFormer-ISTD	0.117	0.095	0.118	0.133

Table 2. Mean road shadow IoU per depth model at the true camera and sun. A useful check should rank DA-V2-L above DA-V2-S; it does the opposite.

5.3. Why: coverage, and agreement at chance

Corollary 2 predicts that compressed depth casts more shadow. We measure this on drive 0001 (every sixth frame, 18 frames, DA-V2-S, threshold detector; Fig. 4). The rendered shadow covers 7.5% of the road at the true depth and 9.9% at γ or $k = 0.6$, 1.3 times as much. The observed shadow covers 31%. Within a frame the IoU tracks the rendered area (correlation 0.48 over all corruption values, against -0.51 for κ), so the wrong depth that casts more shadow scores better.

The more basic finding is the level. At the true depth the IoU is 0.064, and two *independent* masks of the same coverages reach 0.062 (excess 0.002, 95% frame bootstrap [−0.022, 0.034]). The chance-corrected agreement is $\kappa = -0.003$. The rendered shadows carry no information about

the observed ones beyond their area. κ removes the pull towards large masks, but with no signal left it now prefers the *smaller* masks of expanded depth, and the truth is still never the per-frame best. A better score cannot fix this, because there is nothing to score. On the synthetic street, where the observed shadow is exact, κ keeps the signal (Sec. 5.1).

6. Discussion

What a shadow check would need. The theory says what to test and how precisely.

- *Do not claim metric scale.* No shadow score can see it (Proposition 1). Sizes, the camera height, or a second view must supply it.
- *Treat a focal or shape test as a sun test.* By Proposition 2 and Corollary 1, a 20% error is a sun error of a few degrees at most. Before scoring depth, a score should show that it locates the true sun to that precision on single frames. On KITTI it does not even resolve 30°.
- *Use scores that cannot be won by coverage.* IoU against a detector covering a third of the road prefers longer shadows, and compressed depth casts them (Corollary 2). Report the chance level, or a chance-corrected score, next to any overlap score.
- *Choose observable frames.* The equivalent sun error vanishes when the sun is near the optical axis or the image plane (Corollary 1). A per-frame observability weight should come from $\theta(k, \psi)$ before any score is trusted.
- *See the casters.* Most road shadows in KITTI come from trees and facades that are out of view or above the LiDAR. A useful benchmark needs caster–shadow pairs in the image [1], boundary-level matching instead of region overlap, and a renderer that does not assume occluder thickness.

Limitations. All KITTI frames come from one day at one sun elevation (about 40°) in one city. Five drives give wide bootstrap intervals, and some intervals for the compressed-depth gain include zero. The claim that the truth is rarely the argmax holds with tight intervals. The renderer works in screen space at quarter resolution. The coverage and chance analysis uses one drive with the threshold detector. We test four corruptions, one at a time. Real depth errors mix them, and a mixed error is still covered by Proposition 2 when it is diagonal. Our negative result is about cast-shadow *region* scores on driving data. It does not rule out boundary or correspondence scores on scenes with visible casters. It does set the bar such scores have to clear.

7. Conclusion

A known sun makes cast shadows look like a free ruler for monocular depth. They are not a ruler. They cannot see global scale at all, and they see focal length and depth shape only as sun errors of a few degrees. On real driving data, region-overlap scores fail these limits in a predictable way. They reward the longer shadows of compressed depth, and at the true depth they agree with the observed shadows only at chance. The propositions and the corruption protocol give a test that any future shadow-based signal for depth should pass first.

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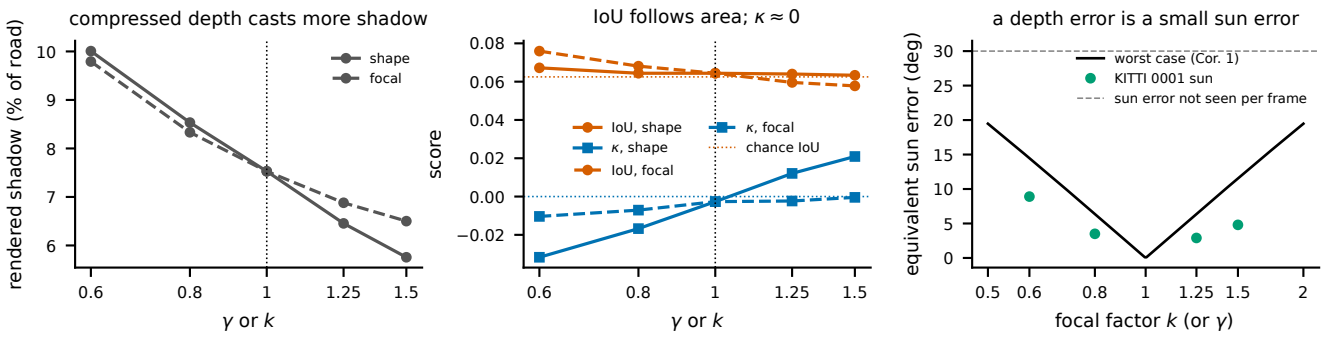


Figure 4. Left: rendered shadow area on KITTI drive 0001 grows as depth is compressed (Corollary 2). Middle: the IoU follows the area and sits on the chance level of independent masks (dotted); κ is near zero. Right: the sun error equivalent to a focal or shape error, the worst case of Corollary 1 and the actual value for the drive's sun. A single KITTI frame does not resolve a 30° sun error (dashed).

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